Visual Diagnostics for NEB Paths

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> *Notes on reading eOn NEB output past the energy profile, with RMSD axes, 2D projections and force tables.*

Background

A smooth nudged elastic band (NEB) energy profile with the climbing image near the top is not enough on its own. It does not say which images were still moving, what the final forces were, or whether the band went the long way round. Those details sit in the run directory's `neb_*.dat` files. A first-order saddle is a critical point, $\nabla E \approx 0$, with an index-1 Hessian (one negative eigenvalue). A smooth profile does not measure the gradient or the index.

The methods come out of the transition-state work on my research site. Labels follow the OCI-NEB paper (Goswami, Gunde, and Jónsson 2026), and the 2D panels follow the RMSD projection paper (Goswami 2026b). The runs reuse the NEB orchestration setup and the Atomistic Cookbook eOn + PET-MAD recipe (Goswami 2026a).

Most of the figures use the 1,3-dipolar cycloaddition of ethylene and nitrous oxide to 4,5-dihydro-1,2,3-oxadiazole. The system is small, but a run still uses the whole stack. eOn (Chill et al. 2014) loads PET-MAD (Mazitov et al. 2025) through metatomic (Bigi et al. 2026). Energy-weighted CI and OCI-NEB run on top. The Grignard and bicyclobutane paths bend away from the straight line, so I leave them for later. Building this stack on Windows needed its own post, in

$$WindowsCompatibilityandScientificComputing$$

().

Reproducing the figures

The companion repository has a pinned environment and a Snakemake workflow. I have not copied the environment file into the post, since a pasted copy goes out of date well before the pinned one does.

```
git clone https://github.com/HaoZeke/nebviz_repro
cd nebviz_repro
pixi run -e cpu dry-run
pixi run -e cpu all
```

The `pixi` tasks call Snakemake under `eonRuns`.

```
cd eonRuns
snakemake --configfile config/general_config.yml -n -p
snakemake --configfile config/general_config.yml -c4 all_figures
```

The `all_figures` rule exports PET-MAD, prepares and minimizes the endpoints, writes the eOn inputs, and runs OCI-NEB for three systems. readcon v2 checks the `.con` files, and the figures are written to `eonRuns/results/03_figures/blog`. The eOn configuration sets `ci_mmf = true` and includes the dimer

block OCI-NEB needs. `stack.json` records `eonclient` 2.14.0 at commit 4a52123, readcon 0.8.0, rgpycrumbs 1.7.0, chemparseplot 1.7.0, NumPy 2.4.2, and matplotlib 3.10.9.

Every plotted input is a plain text file. The final band is `neb.con`, the per-step energies and forces are in `neb_*.dat`, and the path snapshots are `neb_path*.con`. `python -m rgpycrumbs.cli eon plt-neb` draws the profiles and landscapes, and the three table diagnostics read the eOn tables directly.

The blog source also has a smaller bundle under `repro/neb-visualizations/`, which does not rerun any calculation. Its CSV files are the NEB outputs cut down to the columns the figures use, and `pixi run -C repro/neb-visualizations figures` rebuilds the images from them.

One dimension, three coordinates

A standard NEB profile fixes the horizontal axis before you look at it. I plot three versions. Path length is the arc length along the band. Image index lets me compare bands with different numbers of images. The third axis is the root-mean-square deviation (RMSD) from the reactant, as defined in the glossary, computed after alignment with iterative rotations and assignments (IRA). Its unit is Angstroms of atomic displacement.

```
python -m rgpycrumbs.cli eon plt-neb \
  --con-file results/02_neb/system_100/neb.con \
  --output-file results/03_figures/blog/neb-viz-1d-path.png \
  --plot-type profile --rc-mode path --plot-structures crit_points \
  --strip-renderer xyzrender --perspective-tilt 8 --rotation 15x,55y,0z
```

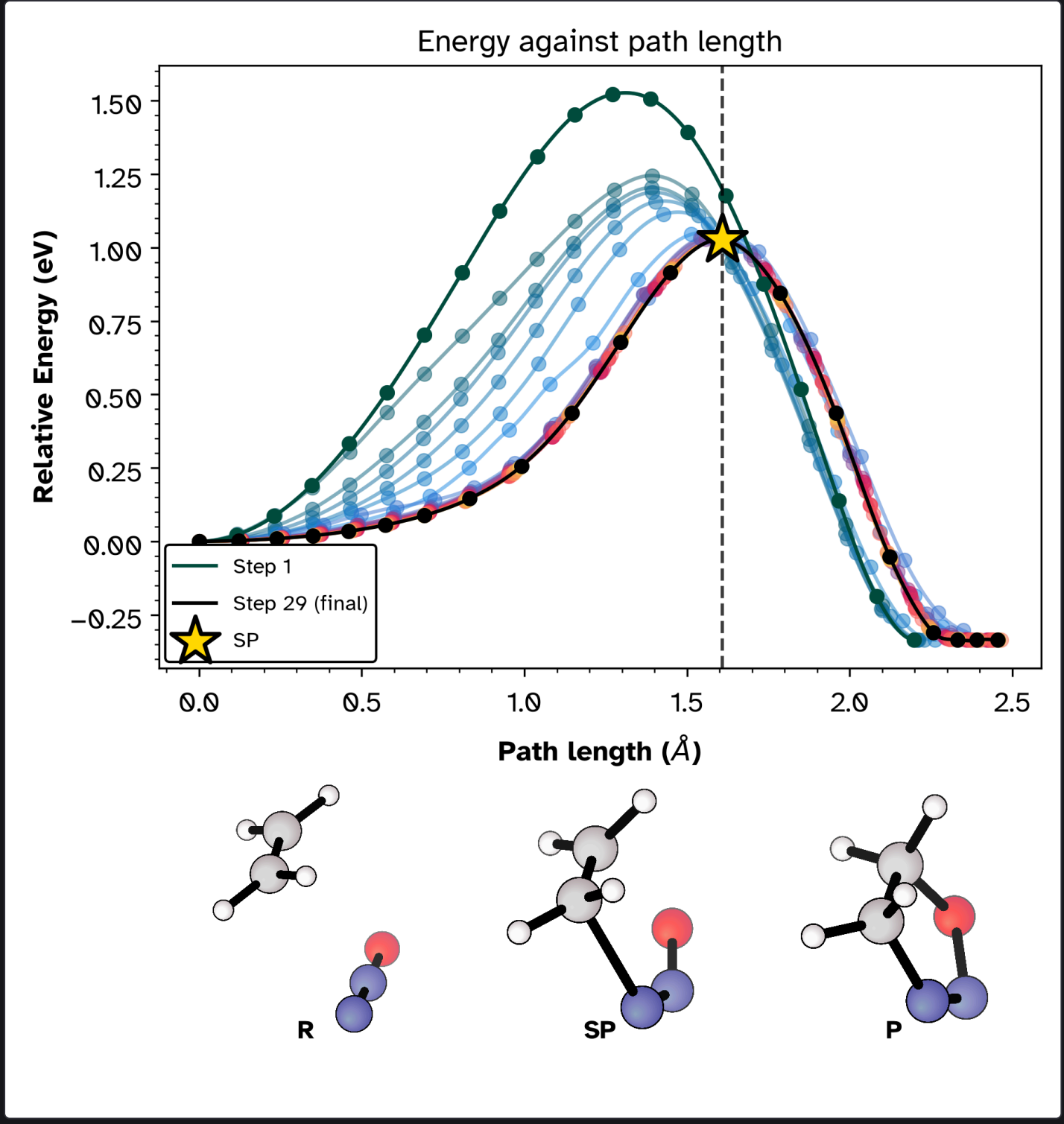


Figure 1: Energy against path length. The peak is the highest-image energy relative to the reactant on this band. The piecewise cubic Hermite spline uses the parallel forces, so the curve matches dE/ds at every image instead of wiggling between points. Dark to light traces the optimization, and the last band is black. R, SP, and P sit in a strip under the panel, off the curve. SP marks the highest image.

For the cycloaddition, the path-length plot is the familiar one. From the initial band to the last one, the highest image drops from about 1.52 eV above the reactant to about 1.03 eV. The product ends about 0.34 eV below the reactant. The max parallel force on this band is still near 2.9 eV/A. The band is not force-converged, and 1.03 eV is only the energy of that highest image. This plot has no way to show motion perpendicular to the band, gaps between images, or a top image in a narrow channel.

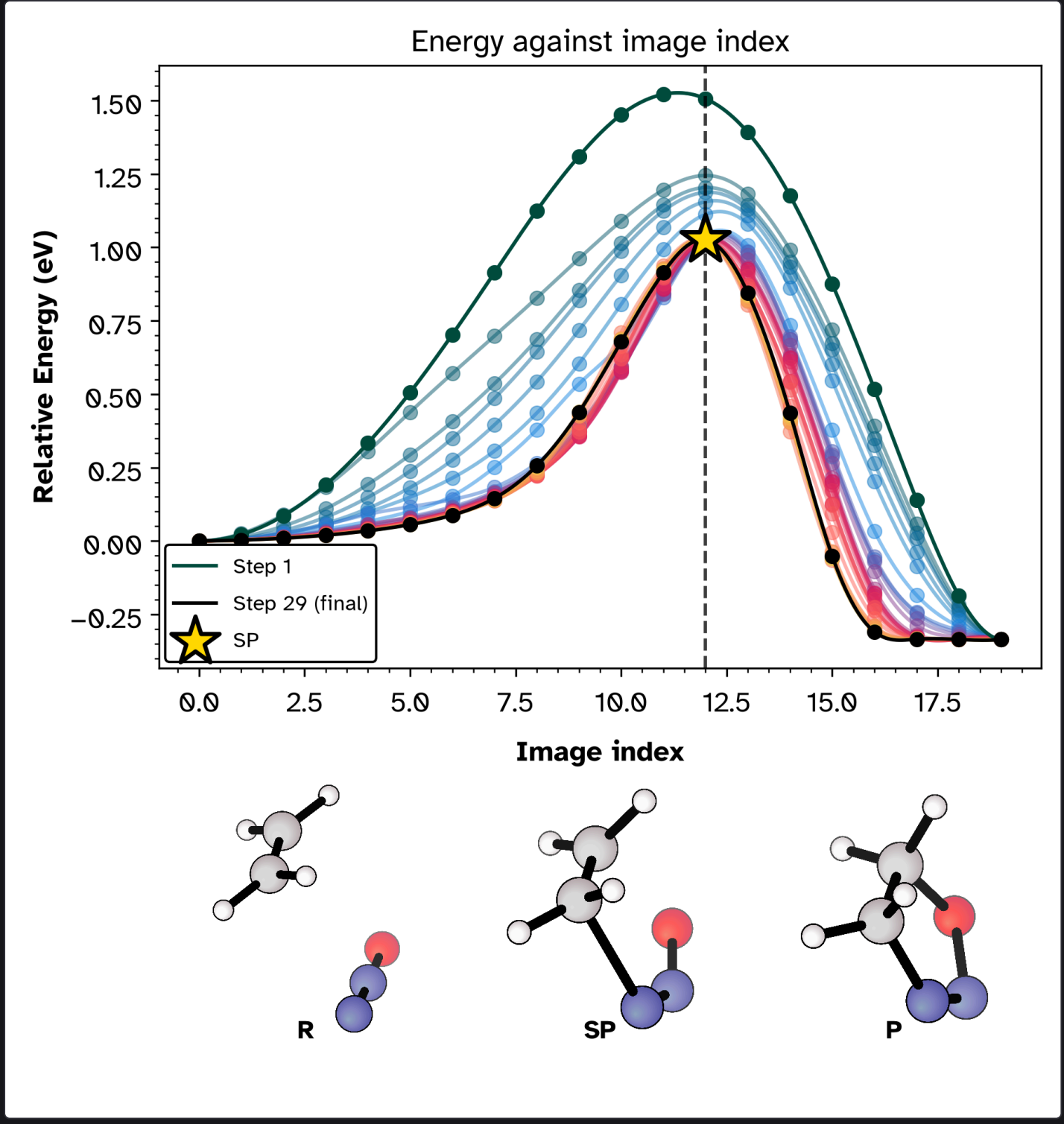


Figure 2: Energy against image index. Even spacing makes bands with different image counts comparable. The curve is a smooth cubic through the energies and not a Hermite spline, since the force slopes are not dE/d(index). Same strip layout as the path-length panel.

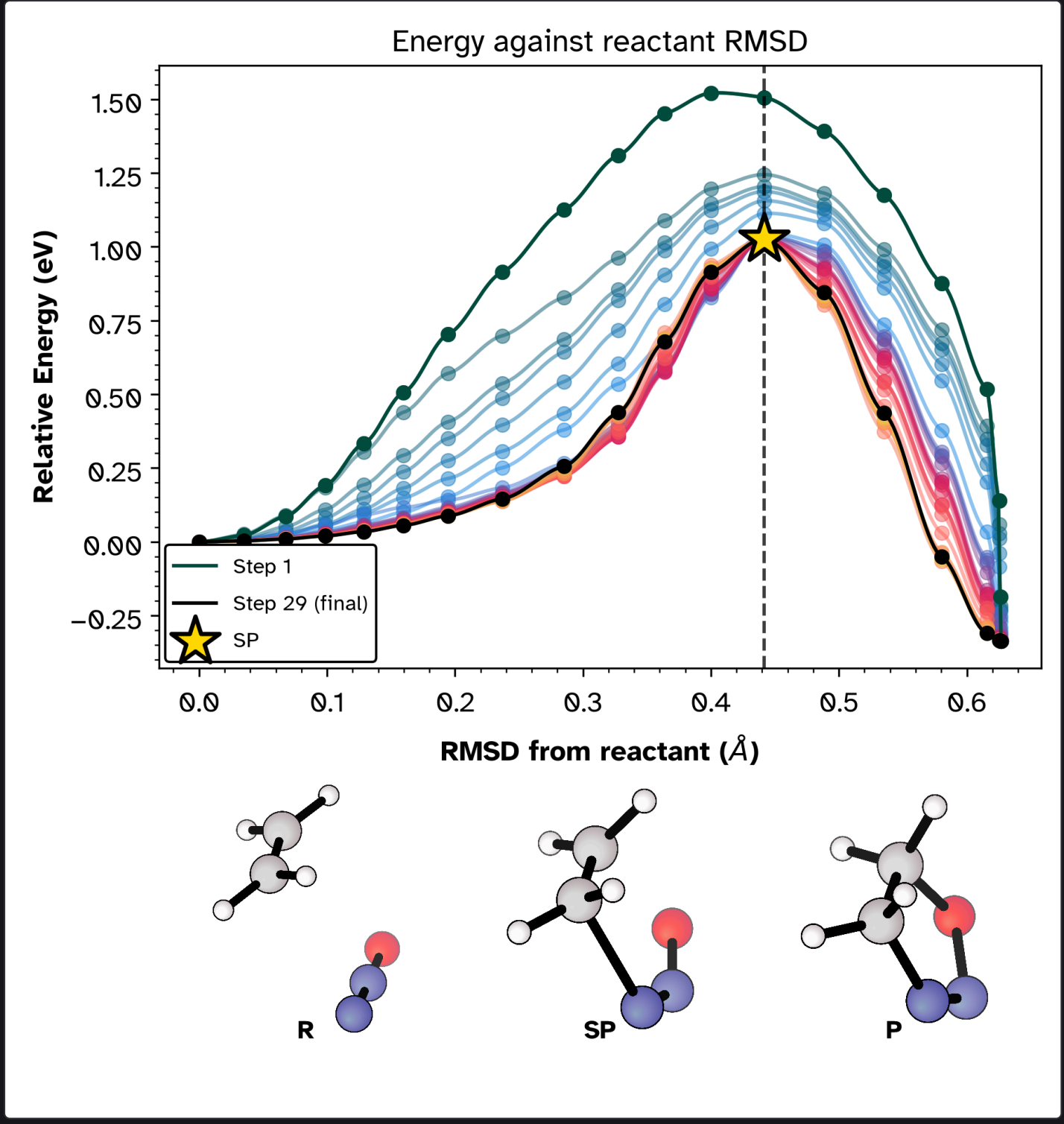


Figure 3: Energy against RMSD from the reactant, so the horizontal coordinate measures Angstroms of real geometric change. Structures stay below the axes to keep the curve readable.

The path-length plot can look smooth while the image-index plot shows images still sliding along the band. The RMSD plot picks up geometric changes that path length absorbs into arc length.

Projection into two dimensions

The 2D projection plots the geometry directly. Each structure gets two IRA RMSD coordinates, r to the reactant and p to the product. I project the NEB parallel forces into the same plane as synthetic gradients. A gradient-enhanced Gaussian process with an inverse-multiquadric kernel is then fit to the energies and gradients together. A rotation then maps r, p to reaction progress s and orthogonal deviation d. The RMSD projection paper (Goswami 2026b) has the derivation and the cases where it fails.

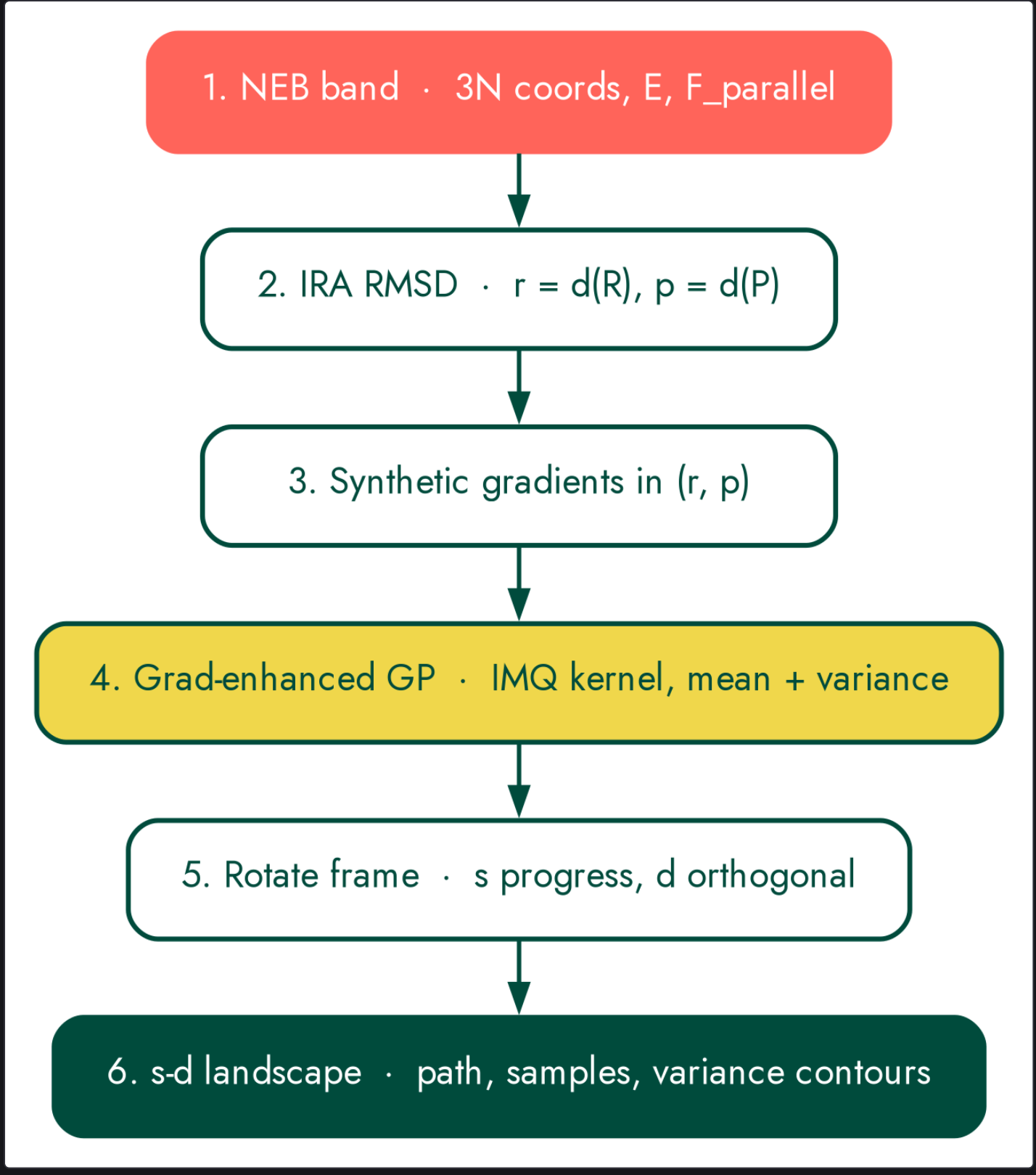


Figure 4: Projection pipeline. The 3N-dimensional NEB output goes to IRA RMSD coordinates, then synthetic gradients, then a gradient-enhanced GP surface, and finally the rotated s-d frame.

```
python -m rgpycrumbs.cli eon plt-neb \
  --con-file results/02_neb/system_100/neb.con \
  --output-file results/03_figures/blog/neb-viz-2d-cyclo.png \
  --plot-type landscape --project-path --show-pts --landscape-path all \
  --plot-structures crit_points --surface-type grad_imq --ira-kmax 14 \
  --title Cycloaddition --theme ruhi --cmap-landscape ruhi_diverging \
  --strip-renderer xyzrender --perspective-tilt 8 --rotation 15x,55y,0z \
  --show-legend --additional-con resources/ewNEB/configurations/system100-sp.xyz \
  ORCA
```

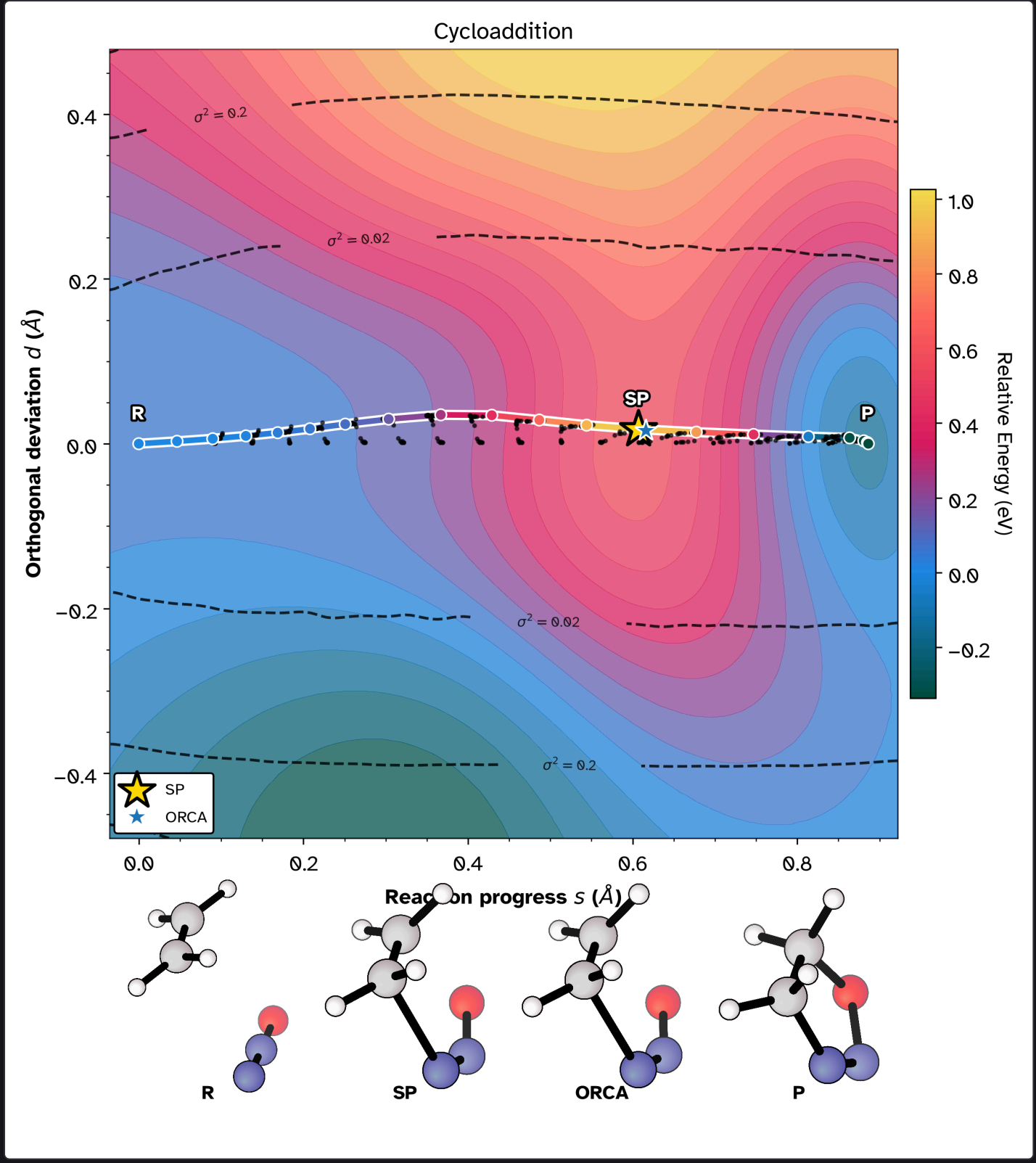


Figure 5: Cycloaddition in the s-d frame. Ruhi energy surface, sampled structures as black dots, the last band as open circles, posterior-variance contours as dashed curves, and the ORCA B3LYP-D3 saddle as a star (strip label: ORCA).

The panel shows the GP energy surface, the sampled structures, the path history, the variance contours, and an external reference saddle. For the cycloaddition the black sample points stay close to the last band, so the optimizer did not leave that channel. The ORCA B3LYP-D3 saddle (Ásgeirsson et al. 2021) comes from an independent calculation and falls in the same part of the s, d plane as the PET-MAD/eOn band. The dashed variance contours are tight near the samples and widen toward the edges, where the surface has few data points. If the highest-image energy and the ORCA barrier differ, the plot shows the offset between the band and that reference. A pair of numbers cannot.

When the path turns

The cycloaddition stays near the diagonal. The Grignard addition (Birkholz and Schlegel 2015) and the bicyclobutane ring opening from the mlFSM work (Marks and Gomes 2025) do not. Both bow away from the reactant-product diagonal in the s, d plane. A 1D profile counts that bend as extra arc length and drops its direction.

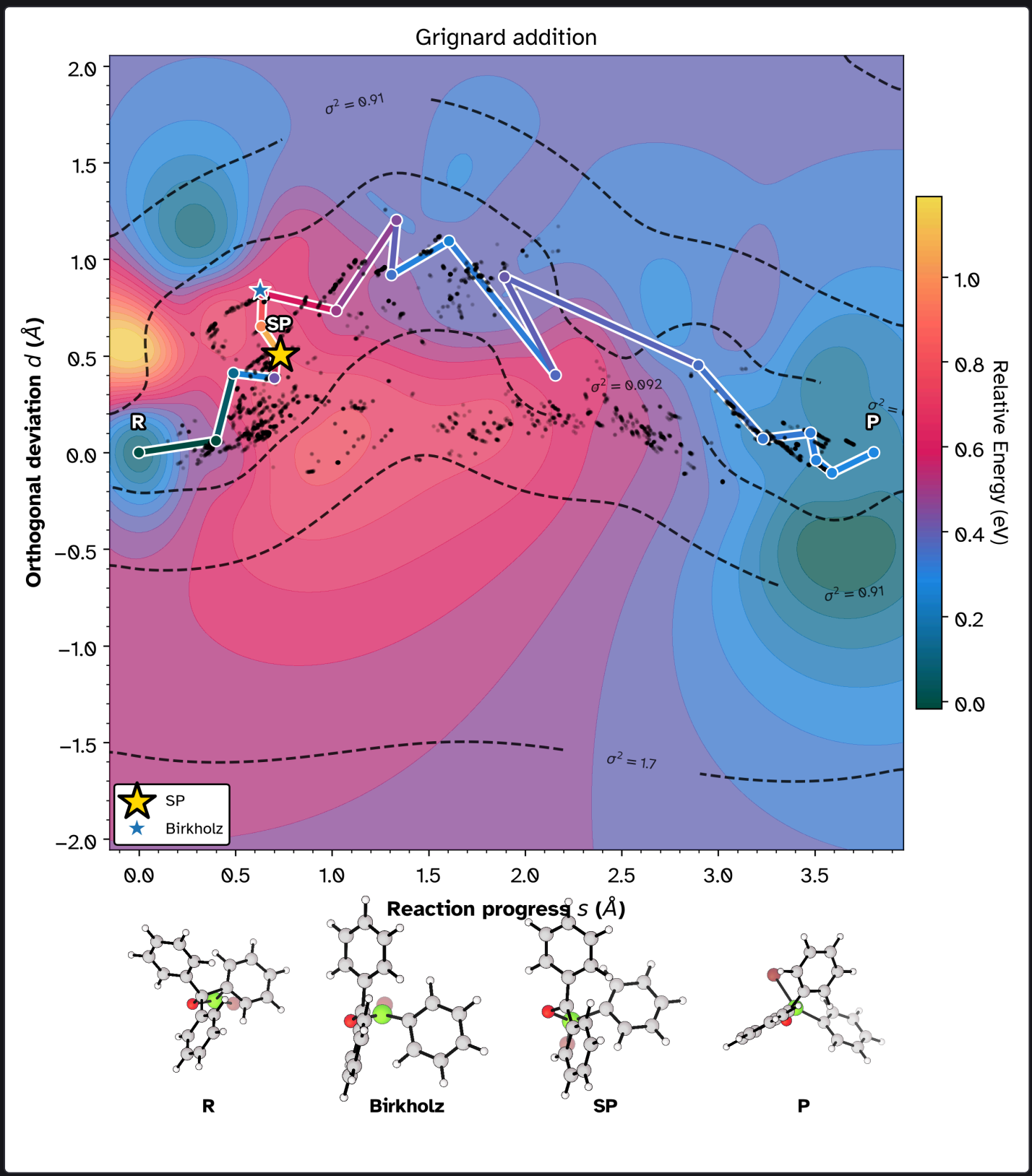


Figure 6: Grignard addition. The band bows off the reactant-to-product line, and Birkholz's reference saddle sits just above and left of the highest image (marked SP). The last band zigzags between s of 1 and 2.2 Angstrom, where the images are far apart (see the spacing plot), and the surface below d of -0.5 has almost no samples, so its colour there is the GP mean rather than data. The 1D coordinate counts the whole bend as arc length.

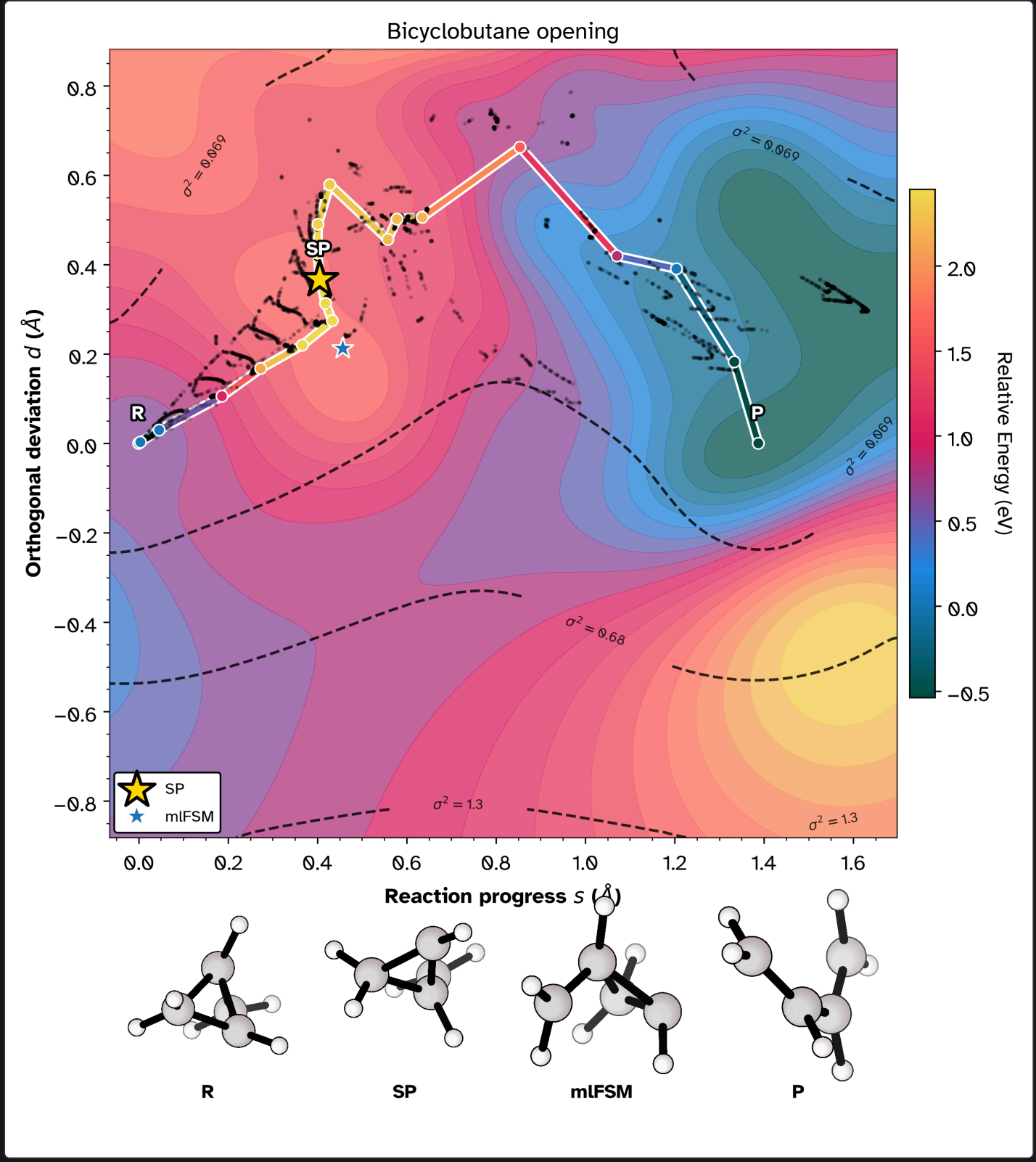


Figure 7: Bicyclobutane ring opening. The kink near s of 0.4 Angstrom shows up as off-diagonal motion. The mlFSM reference saddle sits slightly below the highest image (marked SP). The warm region at the lower right has no samples behind it, so read it off the variance contours, not the colour.

What the force tables say

The tables say whether the band is force-converged. A first-order saddle is a critical point, $\nabla E \approx 0$, with an index-1 Hessian (one negative eigenvalue). A barrier that has stopped changing means the highest-image energy is steady from step to step. That is not $\nabla E \approx 0$, and it says nothing about the Hessian index.

I check the GP surfaces against these tables every time. The Grignard run has force spikes, minimum mode following (MMF) back-off steps, and lost dimer modes. I kept those steps in the figures. Removing them would misrepresent the run.

On the cycloaddition the checks are narrower. Has the maximum parallel force dropped? Which images still carry large parallel forces? How even is the final spacing? The highest-image energy is on the same figure so the flat trace sits next to the forces. The flat trace is not a check.

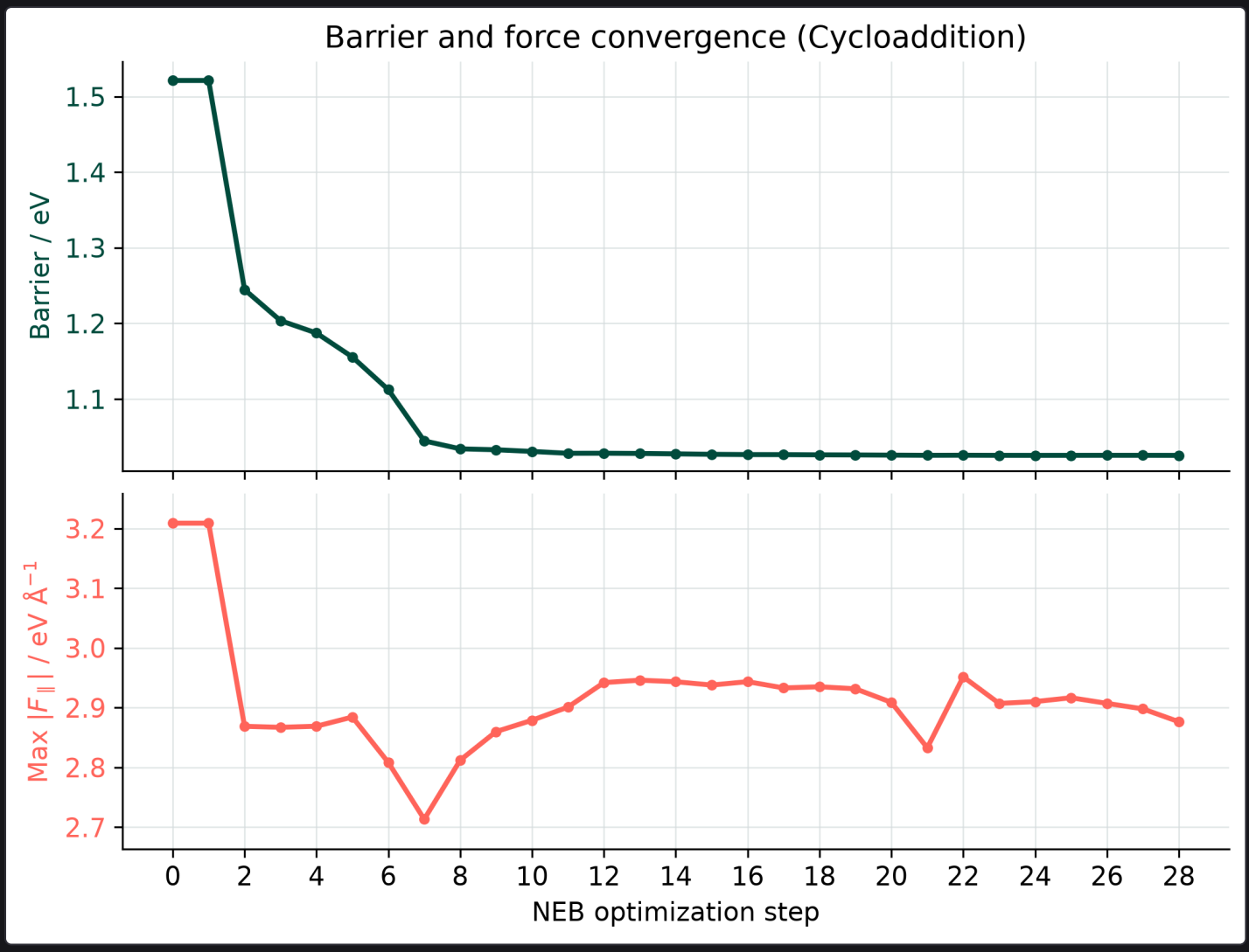


Figure 8: Highest-image energy relative to the reactant (top) and max parallel force (bottom) for the cycloaddition, over the 29 NEB steps (0 to 28). The energy stays near 1.03 eV from step 8 on. The max parallel force never drops below about 2.7 eV/A and ends near 2.9 eV/A. A flat energy trace does not locate a critical point. The band is not force-converged.

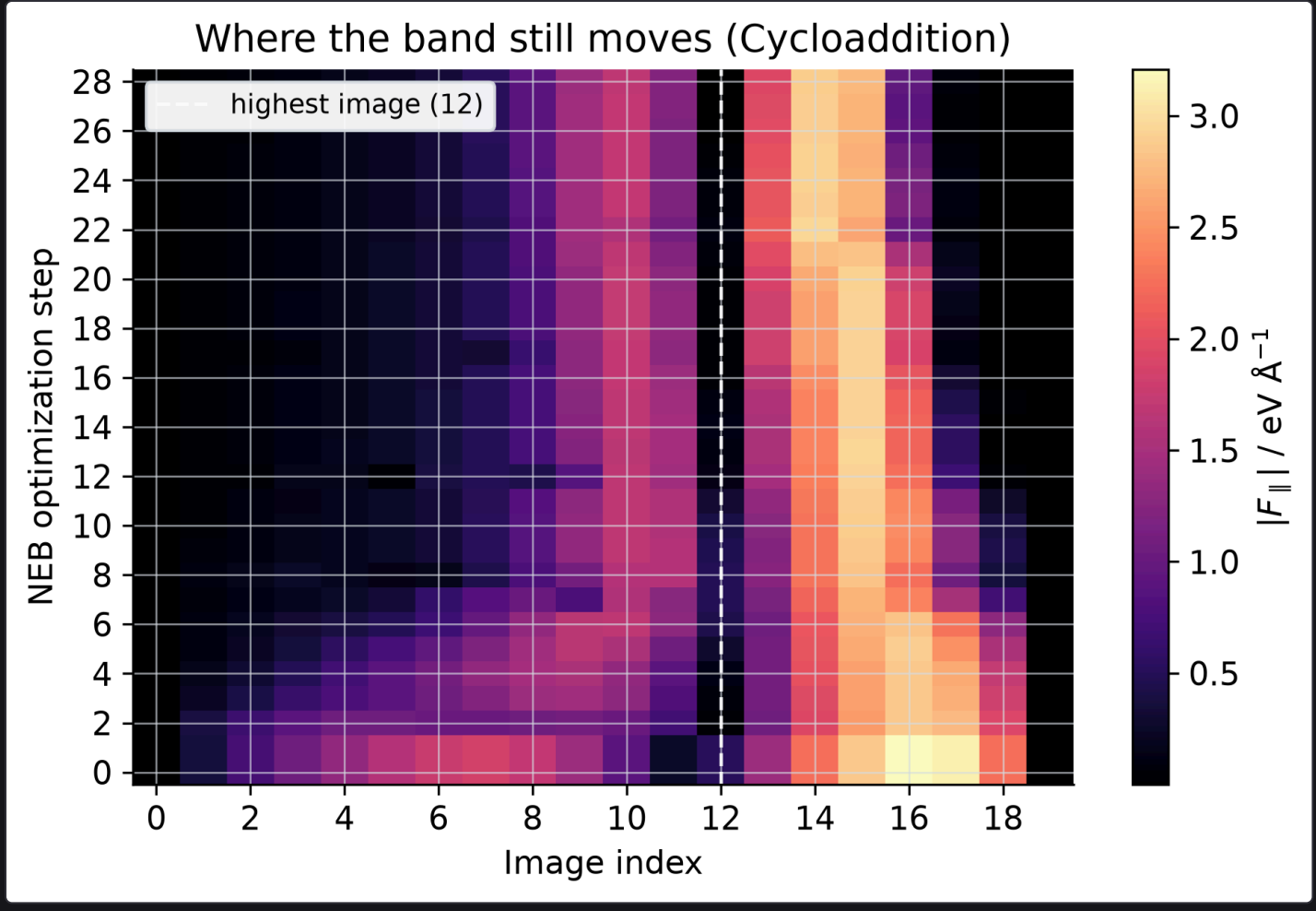


Figure 9: Parallel-force magnitude by image and optimization step for the cycloaddition. The dashed line marks the highest image (12). From step 8 on, the large forces sit on images 13 to 16, on the product side of the climbing image, and not at the top. Those forces are why the band is not force-converged.

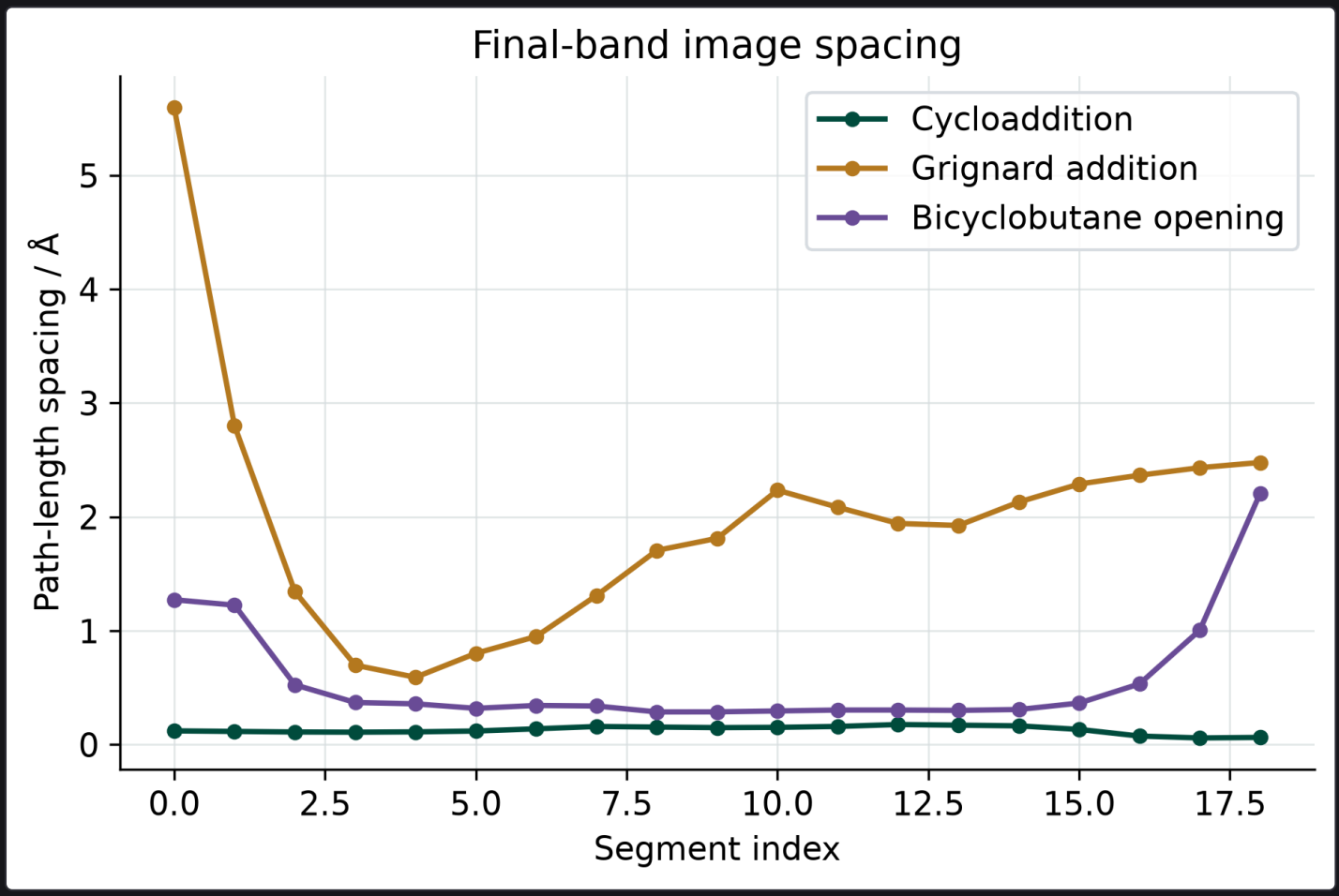


Figure 10: Path-length spacing between neighbouring images on the final band (segment i joins images i and i+1). The cycloaddition is even at about 0.1 Angstrom. The Grignard band has a 5.6 Angstrom first segment and stays above 1.3 Angstrom past segment 7, and the bicyclobutane band opens up at both ends.

Why the figures say OCI-NEB

OCI-NEB is the enhanced climbing image NEB with Hessian eigenmode alignment from (Goswami, Gunde, and Jónsson 2026). Plain CI-NEB (Henkelman, Uberuaga, and Jónsson 2000) pushes the highest image uphill along the band tangent. That reaches the saddle region in few steps. Near the top, though, the tangent is a poor estimate of the unstable mode, and most of the iterations go into that last stretch. Along a low-curvature mode the climbing image can stop moving before it is a first-order saddle, which needs $\nabla E \approx 0$ and an index-1 Hessian.

Once the curvature indicates that the top image is in the transition basin, eOn hands that image to MMF for the saddle search. On the Baker-Chan set with PET-MAD this used 57 percent fewer energy and force calls at the median. The 95 percent credible interval is 50 to 64 percent. On the Pt 111 heptamer set it saved 31 percent across 59 transitions. Switching at a fixed 0.5 eV/A threshold cost 46 percent more calls than the adaptive criterion.

Related publications

- The s, d frame: (Goswami 2026b)
- OCI-NEB and the call-count numbers: (Goswami, Gunde, and Jónsson 2026)
- The Snakemake / eOn lifecycle: (Goswami 2026a)

The images here are built from the reduced tables in `repro/neb-visualizations/`.

Conclusions

On the cycloaddition the highest image levels off near 1.03 eV above the reactant, and the largest parallel force stays near 2.9 eV/A, mostly on images 13 to 16. The band is not force-converged, so 1.03 eV stays the highest-image energy of an unconverged path. A saddle energy needs $\nabla E \approx 0$ and an index-1 Hessian. The RMSD axis and the s, d projection show that the sampled structures stayed in one channel, and that the ORCA saddle sits in the same region. On the Grignard and bicyclobutane bands, the projection and the spacing plot show bends and gaps. The 1D profile records those only as extra arc length. The companion repository regenerates the figures from the eOn output.

Reading list

The 11 references behind these diagnostics, covering the OCI-NEB and RMSD projection papers, the potentials, and the codes, are in a shared citation list.

References

Ásgeirsson, Vilhjálmur, Benedikt Orri Birgisson, Ragnar Bjornsson, Ute Becker, Frank Neese, Christoph Riplinger, and Hannes Jónsson. 2021. "Nudged Elastic Band Method for Molecular Reactions Using Energy-Weighted Springs Combined with Eigenvector Following." *Journal of Chemical Theory and Computation* 17 (8): 4929--45. <https://doi.org/10.1021/acs.jctc.1c00462>.
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Birkholz, Adam B., and H. Bernhard Schlegel. 2015. "Using Bonding to Guide Transition State Optimization." *Journal of Computational Chemistry* 36 (15): 1157--66. <https://doi.org/10.1002/jcc.23910>.
Chill, Samuel T, Matthew Welborn, Rye Terrell, Liang Zhang, Jean-Claude Berthet, Andreas Pedersen, Hannes Jónsson, and Graeme Henkelman. 2014. "EON: Software for Long Time Simulations of Atomic Scale

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---------. 2026b. "Two-Dimensional RMSD Projections for Reaction Path Visualization and Validation." *MethodsX* 16 (March): 103851. <https://doi.org/10.1016/j.mex.2026.103851>.
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Marks, Jonah, and Joseph Gomes. 2025. "Incorporation of Internal Coordinates Interpolation into the Freezing String Method." *Journal of Chemical Theory and Computation* 21 (23). <https://doi.org/10.1021/acs.jctc.5c01492>.
Mazitov, Arslan, Filippo Bigi, Matthias Kellner, Paolo Pegolo, Davide Tisi, Guillaume Fraux, Sergey Pozdnyakov, Philip Loche, and Michele Ceriotti. 2025. "PET-MAD as a Lightweight Universal Interatomic Potential for Advanced Materials Modeling." *Nature Communications* 16 (1): 10653. <https://doi.org/10.1038/s41467-025-65662-7>.